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Positive clutch

Abstract

A plurality of rollers are disposed between an inner race and an outer race that are coaxial and relatively rotatable. The inner race and outer race are configured to engage with each other via part or all of the plurality of rollers that are held in a circumferential direction between roller support parts and pocket parts, the roller support parts being formed on one of an outer circumferential surface of the inner race and an inner circumferential surface of the outer race, and the pocket parts being formed on the other. A torque limiting member that applies a radial load on the rollers to press the rollers against the roller support parts is provided coaxially with the inner race and the outer race. The torque limiting member is flat and annular in an axial direction and resiliently deformable in a radial direction.

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Background/Summary

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present invention relates to a positive clutch equipped with a torque limiter.

2. Description of the Related Art

(2) It is known that some clutches that transmit or interrupt torque from an input shaft to an output shaft are equipped with torque limiters to prevent transmission of excessive torque.

(3) Known torque limiters include frictional torque limiters that use a tolerance ring or frictional plate, or mechanical torque limiters that use, for example, balls and biasing means that applies a force to the balls in the axial or radial direction. Examples of the biasing means include coil springs, disc springs, and plate springs.

(4) Japanese Patent Application Publication No. 2021-156432, for example, discloses a torque limiter, in which a tolerance ring is set either between an outer circumferential surface of an outer race and an inner circumferential surface of a torque input member, or between an outer circumferential surface of an inner race and an outer circumferential surface of a torque input member. When torque exceeding a preset level is applied, the clutch slips between the inner race,

the outer race or the torque input member, and the tolerance ring. The torque limiter thus limits the torque transmission.

(5) Japanese Translation of PCT Application No. 2005-505431 discloses a torque limiter, wherein a circular plate spring is provided to apply a force to balls, which are fitted in holes, to be pushed toward one side. When an excessive torque is applied, for example, the plate spring radially expands through resilient deformation, pressing the balls out of respective holes thereof. This action serves to limit the torque transmission.

SUMMARY OF THE INVENTION

(6) Many frictional torque limiters, including the one described in Japanese Patent Application Publication No. 2021-156432, rely heavily on the friction coefficient of the friction plate. With a friction plate making sliding contact when the clutch is tripped (during torque transmission interruption), with a clutch equipped with such a torque limiter it is difficult to control the torque limit load in response to changes in ambient conditions. After repeated tripping, wear may compromise durability and the limit load may become unstable.

(7) Mechanical torque limiters that use biasing means such as plate springs, including the one described in Japanese Translation of PCT Application No. 2005-505431, cannot easily generate high torque and necessitate a considerable installation space for the biasing means. Therefore, these torque limiters are not well-suited for power transmission applications in vehicles, for which high torque and space saving are required. Other issues that have to be addressed include an increased number of components and higher production costs. Moreover, the torque limiter requires positioning of the plate spring.

(8) The present invention has been made based on the circumstances described above and it is an object of the present invention to provide a positive clutch with a simple structure, exhibiting high durability against wear, excellent sliding characteristics, size reduction capability, extended service life, and the capacity to handle high torque.

(9) The present invention achieves the above object by providing a positive clutch including: an inner race and an outer race that are coaxial and rotatable relative to each other; and a plurality of rollers disposed between the inner race and the outer race and transmitting or interrupting torque, the inner race and the outer race being configured to engage with each other via part or all of the plurality of rollers held in a circumferential direction between roller support parts and pocket parts, the roller support parts being formed on one of an outer circumferential surface of the inner race and an inner circumferential surface of the outer race, and the pocket parts being formed on the other of the outer circumferential surface of the inner race and the inner circumferential surface of the outer race and being configured to accommodate the rollers. The positive clutch further includes a torque limiting member that applies a radial load on the rollers to press the rollers against the roller support parts, the torque limiting member being coaxial with the inner race and the outer race, and being flat and annular in an axial direction, and moreover being resiliently deformable in a radial direction.

(10) According to one aspect of embodiments, the torque limiting member is capable of applying a large radial preload. The axially flat, annular structure allows the torque limiting member to be installed in a confined space. Therefore, the clutch is capable of handling high torque with a simple structure, and can be made smaller.

(11) The torque limit load can be set based on the resilience of the torque limiting member and the friction between the sliding surfaces of the rollers and the pocket parts. That is, the torque limit load can readily be tailored without dependence on friction coefficients. This helps reduce potential wear on the sliding parts and improve durability. The clutch can thus achieve both compactness and performance stability.

(12) Moreover, the rollers that are engaging elements, via which the inner race and the outer race engage, are used as components for causing resilient deformation of the torque limiting member. This enables a reduction in the number of components, which in turn facilitates cost savings, size

reduction, and structural simplification.

(13) According to another aspect of embodiments, the torque limiting member is a spiral retaining ring, which is circumferentially continuous, so that there is no need for phase matching during its assembly. The number of turns of coiling of the spiral retaining ring may be increased to help disperse the stress applied to the ring, ensure strength, and improve durability.

(14) According to another aspect of embodiments, the rollers each have an outer circumferential surface serving as a retaining ring receiving surface. The number of components is thus reduced, which facilitates structural simplification and cost reduction. When the clutch is tripped, the rollers roll between the torque limiting member, inner race, and outer race. Therefore, durability against wear is improved.

(15) Since the torque limiting member is provided so as to apply a load to both axial ends of each roller, the rollers are pressed down in a stable manner, which can contribute to better performance stability. The torque limiting member may be provided such that its outer circumferential surface makes contact with the retaining ring receiving surfaces of the rollers. In this case, the centrifugal force during high-speed rotation assists the loading and enables the clutch to handle high torque.

(16) According to another aspect of embodiments, there is an angle difference between a wall surface of the roller support part and a wall surface of the pocket part that hold each of the rollers. This angle difference allows for change of the proportion of the torque limit load carried by the friction between the sliding surfaces of the rollers and the pocket parts, and the torque limit load carried by the resilience of the torque limiting member. This helps improve performance stability. Moreover, it is possible to customize the trip torque based on the rotating direction of one of the inner race and outer race relative to the other.

(17) According to another aspect of embodiments, the positive clutch is configured as a selectable clutch that can switch operating modes thereof. This clutch includes a torque limiter with a piston mechanism that uses the torque limiting member described above. The clutch can thus achieve both compactness and performance stability. The torque limit load can be changed based on whether the rotating direction is forward or reverse. Moreover, the torque limiting function may be provided for only one of the forward and reverse directions. This offers a high degree of design freedom.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is an exploded perspective view illustrating a configuration example of a positive clutch according to a first embodiment of the present invention;

(2) FIG. 2 is a plan view from the front in the axial direction illustrating the configuration of the positive clutch shown in FIG. 1 with some parts omitted;

(3) FIG. 3 is a cross-sectional view illustrating a part of the section A-A in FIG. 2;

(4) FIG. 4 is an enlarged view of the region encircled with a broken line in FIG. 2;

(5) FIG. 5 is a vector diagram for explaining the function of the torque limiter in the positive clutch shown in FIG. 1;

(6) FIG. 6A is a diagram illustrating a state of a torque limiting mechanism activated upon application of torque exceeding a preset level to the inner race in the forward direction;

(7) FIG. 6B is a diagram illustrating a state in which the inner race and outer race are disengaged and freed from the torque transmission;

(8) FIG. 7 is a cross-sectional view along the rotation axis illustrating a part of another configuration example of the positive clutch according to the first embodiment of the present invention;

(9) FIG. 8 is a cross-sectional view along the rotation axis illustrating a part of yet another configuration example of the positive clutch according to the first embodiment of the present

invention;

(10) FIG. **9** is a cross-sectional view along the rotation axis illustrating a part of a configuration example of a positive clutch according to a second embodiment of the present invention;

(11) FIG. **10** is a plan view from the front in the axial direction illustrating essential parts of the positive clutch shown in FIG. **9** to a larger scale;

(12) FIG. **11** is an exploded perspective view illustrating a configuration example of a positive clutch according to a third embodiment of the present invention;

(13) FIG. **12** is a cross-sectional view of a plane along the rotation axis illustrating the configuration of the positive clutch shown in FIG. **11**;

(14) FIG. **13** is a cross-sectional view along the line B-B in FIG. **12**;

(15) FIG. **14** is an enlarged cross-sectional view illustrating a configuration of essential parts of the positive clutch;

(16) FIG. **15** is a perspective view illustrating the configuration of a switching member;

(17) FIG. **16A** is a diagram illustrating a state of a torque limiting mechanism activated upon application of torque exceeding a preset level to the inner race in the forward direction;

(18) FIG. **16B** is a diagram illustrating a state in which the inner race and outer race are disengaged and freed from the torque transmission;

(19) FIG. **17A** is a side view illustrating a state of the positive clutch shown in FIG. **11** when the operating mode is a one-way lock mode;

(20) FIG. **17B** is a schematic diagram illustrating the state of a first roller and a second roller when the operating mode of the positive clutch shown in FIG. **11** is the one-way lock mode;

(21) FIG. **18A** is a side view illustrating a state of the positive clutch shown in FIG. **11** when the operating mode is a two-way free mode; and

(22) FIG. **18B** is a schematic diagram illustrating the state of the first roller and second roller when the operating mode of the positive clutch shown in FIG. **11** is the two-way free mode.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

(23) Hereinafter, embodiments of the present invention will be described with reference to the drawings.

First Embodiment

(24) As shown in FIG. **1** to FIG. **4**, the positive clutch **100** according to the first embodiment includes an inner race **110** and an outer race **115** rotatable about the same axis relative to each other, a plurality of rollers **130** disposed between the inner race **110** and the outer race **115** for transmission or interruption of torque, and side plates **140**, one each fixed to the front side and rear side of the inner race **110** with screws, for example, and covering the front side and rear side in the axial direction of the inner race. Reference symbol C in FIG. **1** to FIG. **3** denotes the rotation axis.

(25) The inner race **110** is a cylindrical member with roller support parts **121** formed at predetermined circumferential intervals on the outer circumferential surface for supporting the lower side of the rollers **130**. The roller support parts **121** are formed as recessed grooves with a circular arc cross-sectional shape, for example, extending in the direction of the rotation axis. The opening edge of the roller support part **121** is chamfered, for example, as viewed in cross section, which helps facilitate the rolling of the rollers **130** and reduce the ratchet noise.

(26) The outer race **115** is an annular plate member with pocket parts **125** configured to accommodate the rollers **130** and formed at predetermined circumferential intervals on the inner circumferential surface, each corresponding to each of the roller support parts **121**. The pocket parts **125** are formed as recessed grooves extending in the direction of the rotation axis and having a cross-sectional shape that is formed by an isosceles trapezoid and a circle that passes through the vertices of the two equal sides, for example.

(27) The rollers **130** are columnar, accommodated in the pocket parts **125** of the outer race **115** such as to be rollable, and supported by the roller support parts **121** of the inner race **110**. The positive clutch **100** of this embodiment is configured such that the inner race **110** and outer race **115** engage

via the rollers **130**, which are held between the roller support parts **121** and the pocket parts **125** in the circumferential direction when one of the inner race **110** and outer race **115** is rotated relative to the other in the forward or reverse direction. Therefore, no windup or torsional deflection occurs during engagement. The clutch thus has high rigidity and is capable of stable engagement with a simple structure. With a large number of rollers **130** accommodated in a small space, the clutch is able to transmit high torque despite the small size. The large number of rollers also enables a reduction in the surface pressure that acts on the roller support parts **121** and pocket parts **125** as well as on the rollers **130** during engagement. This allows the clutch to be designed with inexpensive materials that are more resistant to chipping, which may be caused by impact, or wear. The rollers **130** rotating themselves and substantially not engaging at the same point allow the durability of the clutch to be improved and the service life to be extended.

(28) The positive clutch **100** according to this embodiment includes ring-like torque limiting members **150** that are resiliently deformable in the radial direction, flat in the axial direction, and coaxial with the inner race **110** and outer race **115**, for applying a radial load on each of the rollers **130**.

(29) The torque limiting member **150** in this embodiment is a spiral retaining ring made by coiling flat wire into a ring-like shape that lies flat in the axial direction. This embodiment uses a spiral retaining ring with two turns of flat wire.

(30) In the positive clutch **100** of this embodiment, the outer circumferential surface of each roller **130** serves as a retaining ring receiving surface. The torque limiting members **150** are fitted on the front side and the rear side of the outer race **115** such that their inner circumferential surfaces make contact with the respective retaining ring receiving surfaces of the rollers **130**. The rollers **130** are each loaded radially at both axial ends so that the rollers **130** are pressed against their respective roller support parts **121**.

(31) The outer circumferential surface of each roller **130** serving as a retaining ring receiving surface helps reduce the number of components, facilitating structural simplification as well as cost reduction. When the clutch is tripped, the rollers **130** roll between the torque limiting members **150**, inner race **110**, and outer race **115**. Therefore, durability against wear is improved.

(32) Since the torque limiting members **150** are provided such as to apply a load to both axial ends of each roller **130**, the rollers **130** are pressed down in a stable manner, which can contribute to better performance stability.

(33) As shown in FIG. 5, when the inner race **110** and outer race **115** engage, the roller **130** receives a normal force T_b [N] expressed as $(T_r \cdot \sin \theta_1)$ from the wall surface of the roller support part **121** of the inner race, where T_r [N] is the load acting on the roller **130**. Angle θ_1 represents the angle $[\circ]$ expressed as $(90^\circ - \theta_n)$, where θ_n is the angle $[\circ]$ of the load acting surface of the wall of the roller support part **121** holding the roller **130** relative to the radial direction, as shown in FIG. 4.

(34) The pocket part **125** receives a component of force T_c [N] expressed as $(T_b \cdot \cos \theta_2)$ from the roller **130** parallel to the sliding surface in the direction along the wall surface of the pocket part **125**, as well as a normal force T_d [N] expressed as $(T_b \cdot \sin \theta_2)$ perpendicular to the sliding surface between the roller **130** and the pocket part **125**. Angle θ_2 represents the angle $[\circ]$ expressed as $(90^\circ - (\theta_n - \theta_g))$, where θ_g is the angle $[\circ]$ of the sliding surface relative to the radial direction, of the wall surface of the pocket part **125** holding the roller **130**, as shown in FIG. 4. Furthermore, a frictional force T_f [N] expressed as (μT_d) acts on the sliding surface between the roller **130** and the pocket part **125**, where μ represents the friction coefficient of the sliding surface.

(35) The torque limiting member **150** receives a pressing force N_c [N] expressed as $(T_c - \mu T_d)$ from the roller **130** in the direction along the wall surface of the pocket part **125**. Namely, the torque limiting member **150** is subjected to a radial pressing force N [N] expressed as $(N_c \cdot \cos \theta_2)$.

(36) When the input torque is so large as to cause the radial pressing force N to exceed the resilience of the torque limiting member **150**, i.e., the radial load applied on the roller **130**, the rollers **130** roll and move toward the bottom of the pocket parts **125**. This movement allows the

torque limiting member **150** to radially expand through resilient deformation as shown in FIG. 6A. As the rollers **130** roll out of the roller support parts **121** onto the outer circumferential surface of the inner race **110** as shown in FIG. 6B, the inner race **110** and outer race **115** disengage and are freed from the torque transmission. This limits the magnitude of torque transmitted between the inner race **110** and the outer race **115**.

(37) In this embodiment, as shown in FIG. 4, there is a difference between the angle θ_n of the load acting surface of the roller support part **121** and the angle θ_g of the sliding surface of the pocket part **125**. This angle difference allows for change of the proportion of the torque limit load carried by the frictional force T_f on the sliding surface between the roller **130** and the pocket part **125**, and the torque limit load carried by the resilience of the torque limiting member **150**. This feature helps improve performance stability. Moreover, it is possible to customize the trip torque based on the rotating direction of one of the inner race **110** and outer race **115** relative to the other.

(38) In the positive clutch **100** described above, the torque limiting member **150** is a spiral retaining ring. The number of turns of coiling can be suitably changed, for example, to apply a high radial preload. The ring allows for installation in a confined space. Therefore, the clutch is capable of handling high torque with a simple structure, and can be made smaller.

(39) Since spiral retaining rings are circumferentially continuous, there is no need for phase matching when assembling the torque limiting member **150**. Increasing the number of turns of coiling can help disperse the stress applied to the torque limiting member **150**, ensure strength, and improve durability.

(40) The torque limit load can be set based on the resilience of the torque limiting member **150** and the friction between the sliding surfaces of the rollers **130** and the pocket parts **125**. That is, the torque limit load can readily be tailored without dependence on friction coefficients. This helps reduce potential wear on the sliding parts and improve durability. The clutch can thus achieve both compactness and performance stability.

(41) Moreover, the rollers **130** are used as components for causing resilient deformation of the torque limiting members **150**. This enables a reduction in the number of components, which in turn facilitates cost savings, size reduction, and structural simplification.

(42) In the positive clutch **100**, as described in the first embodiment above, the torque limiting member **150** is a spiral retaining ring made of two turns of flat wire. The number of turns and other features of the spiral retaining ring are not particularly limited and may be set suitably to generate a resilient force matching the necessary torque load. For example, a spiral retaining ring of three turns of flat wire as shown in FIG. 7 may be used as the torque limiting member **150**.

(43) In the positive clutch **100** according to the first embodiment described above, torque limiting members **150** are provided such as to apply a load to both axial ends of each of the rollers **130**. Instead, a torque limiting member **150** may be provided such as to apply a load to one axial end of each of the rollers **130**, as shown in FIG. 8.

(44) In the positive clutch **100** according to the first embodiment described above, the torque limiting member **150** is a spiral retaining ring. Instead, the torque limiting member **150** may be a plurality of C-shaped retaining rings, for example, stacked in the axial direction.

(45) In the positive clutch according to the first embodiment described above, the torque limiting member **150** is fitted on the rollers such that the inner circumferential surface thereof makes contact with the outer circumferential surface of each of the rollers **130** serving as the retaining ring receiving surface. Instead, the torque limiting member **150** may be fitted inside the rollers so that the outer circumferential surface thereof makes contact with the outer circumferential surface of each of the rollers **130** serving as the retaining ring receiving surface.

Second Embodiment

(46) FIG. 9 is a cross-sectional view along the rotation axis illustrating a part of a configuration example of a positive clutch according to a second embodiment of the present invention. FIG. 10 is a plan view from the front in the axial direction illustrating essential parts of the positive clutch

shown in FIG. 9 to a larger scale.

(47) In this positive clutch, the roller support parts **121** are formed on the inner circumferential surface of the outer race **115**, and the pocket parts **125** are formed on the outer circumferential surface of the inner race **110**.

(48) The torque limiting members **150** are fitted on the front side and the rear side of the inner race **110** such that their outer circumferential surfaces make contact with the respective retaining ring receiving surfaces of the rollers **130**. The torque limiting members thus apply a radial load to both axial ends of each roller **130**, causing the rollers **130** to be pressed against their respective roller support parts **121**. In this embodiment, the torque limiting members **150** are two C-shaped retaining rings, for example, stacked in the axial direction. Instead, the torque limiting members **150** may be spiral retaining rings of two turns of flat wire.

(49) In this positive clutch, when the input torque is so large as to cause the radial pressing force of the rollers **130** to exceed the resilience of the torque limiting members **150**, the rollers **130** roll and move toward the bottom of the pocket parts to allow the torque limiting members **150** to radially shrink through resilient deformation. As the rollers **130** roll out of the roller support parts **121** onto the inner circumferential surface of the outer race **115**, the inner race **110** and outer race **115** disengage and are freed from the torque transmission. This limits the magnitude of torque transmitted between the inner race **110** and the outer race **115**.

(50) The positive clutch according to the second embodiment provides the same effects as those of the positive clutch **100** of the first embodiment. Moreover, the centrifugal force during high-speed rotation assists the loading, which ensures more stable engagement and allows the clutch to handle high torque.

Third Embodiment

(51) FIG. **11** is an exploded perspective view illustrating a configuration example of a positive clutch according to a third embodiment of the present invention.

(52) FIG. **12** is a cross-sectional view of a plane along the rotation axis illustrating the configuration of the positive clutch shown in FIG. **11**. FIG. **13** is a cross-sectional view along the line B-B in FIG. **12**. FIG. **14** is an enlarged cross-sectional view illustrating a configuration of essential parts of the positive clutch.

(53) The positive clutch **100** according to the third embodiment is configured as a selectable clutch that allows switching of the operating modes. The clutch includes an inner race **110** and an outer race **115** rotatable about the same axis relative to each other, a plurality of rollers **130** disposed between the inner race **110** and the outer race **115** for transmission or interruption of torque, a biasing means **160** biasing each of the rollers **130** in the radial direction, a side plate **140** fixed to the rear side of the outer race **115** with screws, for example, and covering the rear side in the axial direction of the outer race, a switching member **170** for switching the operating modes of the positive clutch **100**, and a torque limiter **151** configured with a piston mechanism that uses a torque limiting member **150**. Reference symbol C in FIG. **11** and FIG. **12** denotes the rotation axis.

(54) The inner race **110** has roller support parts **121** formed at predetermined circumferential intervals on the outer circumferential surface for supporting the lower side of the rollers **130**.

(55) In this embodiment, two rollers **130**, each held in each of two adjacent roller support parts **121**, make a pair. Four roller pairs **131** are arranged circumferentially at equal intervals.

Hereinafter, for convenience of explanation, the clockwise direction in FIG. **13** shall be referred to as the forward direction, and the counterclockwise as the reverse direction. One of the rollers **130** of the roller pair **131** located on the side in the forward direction will be referred to as the first roller **130a**, and the other of the rollers **130** on the side in the reverse direction as the second roller **130b**. Unless otherwise specified, the first roller **130a** and second roller **130b** will be referred to simply as roller(s) **130**.

(56) The number of roller pairs **131** is not limited to a particular number, and may be changed as required corresponding to a target torque capacity. The roller pairs **131** need not be equally spaced

apart.

(57) In the inner circumferential surface of the outer race **115** are formed first pocket parts **125a** corresponding to the first rollers **130a** of the roller pairs **131**, extending at an angle in the forward direction radially outwards. Second pocket parts **125b** corresponding to the second rollers **130b** of the roller pairs **131** extend at an angle in the reverse direction radially outwards.

(58) The first pocket parts **125a** and second pocket parts **125b** are configured to be able to accommodate one roller **130** each inside to allow the rollers **130** to separate from the inner race **110**.

(59) The biasing means **160** is a garter spring, for example. The rollers **130** each have a circumferentially extending mounting groove **135** on the outer circumferential surface. The biasing means is mounted in these mounting grooves. Using a garter spring as the biasing means **160** eliminates the need to provide multiple biasing means corresponding to each of the rollers **130**. This reduces the number of components and helps reduce the size of the clutch.

(60) The switching member **170** is configured to be moved in the axial direction independently of the rotation of the inner race **110** and outer race **115**. This axial movement causes one or both of the first rollers **130a** and second rollers **130b** to move into the corresponding pocket parts, to allow the positive clutch **100** to be switched between operating modes. The switching member **170** includes, as shown in FIG. 15, an annular plate-like base part **171**, and a plurality of roller position changing parts **172** integrally formed on the front face of the base part **171** to extend in the axial direction, each corresponding to each of the roller pairs **131**.

(61) Each roller position changing part **172** has a pillar part **173**, a first pressing part **174** on one side in the forward direction of the pillar part **173** for pressing the first roller **130a**, and a second pressing part **175** on the side in the reverse direction of the pillar part **173** for pressing the second roller **130b**.

(62) The first pressing part **174** is located further towards the rear in the axial direction than the second pressing part **175**. By tailoring the amount of axial movement of the switching member **170**, it is possible to change the positions of the second rollers **130b** alone of the roller pairs **131**, or the positions of both of the first rollers **130a** and second rollers **130b**.

(63) The roller position changing parts **172** are each passed through respective through holes **141** formed in the side plate **140**. The pillar part **173** of each roller position changing part is positioned radially close to the inner race **110** between the first roller **130a** and the second roller **130b** of the corresponding roller pair **131**.

(64) The distal ends of the roller position changing parts **172** pass through and extend out from through holes **127** in the outer race **115**. An annular plate-like retainer member **176** is fixed to the front end face of each roller position changing part with screws, for example.

(65) The torque limiter **151** includes piston members **155**, and the torque limiting member **150** in contact with the piston members **155** and applying a constant load on the piston members **155**. The piston member **155** is a columnar body that is formed by cutting one end of the column in a plane inclined to the center axis.

(66) In this embodiment, as shown in FIG. 14, the torque limiter **151** has the piston members each provided to the second roller **130b** in each roller pair **131** and located on one side in the forward direction of the second roller **130b**. Piston member accommodating holes **126** are formed in the wall surface on the radially outer side of the second pocket parts **125b** of the outer race **115**. The piston members **155** are slidably accommodated in the holes, with one end protruding beyond the outer circumferential surface of the outer race **115**. The piston members **155** are oriented such that their center axis is inclined toward the forward direction relative to the radial direction. The second roller **130b** can be held in the circumferential direction between the flat surface at the distal end of the piston member **155** and the roller support part **121**.

(67) In this embodiment, the torque limiting member **150** is two spiral retaining rings of two turns of flat wire stacked in the axial direction. The torque limiting member **150** is fitted on the rear side

of the outer race **115** such that the inner circumferential surface makes contact with each of the piston members **155**. The torque limiting member thus applies a radial load to press the rollers **130** against their respective roller support parts **121**.

(68) In the positive clutch **100** according to this embodiment, when the switching member **170** is fixed at the first fixed position where the retainer member **176** is in contact with the front face of the outer race **115** as shown in FIG. **12**, the inner race **110** and outer race **115** engage via the second rollers **130b**, which are held between the roller support parts **121** and the piston members **155** in the circumferential direction when the inner race **110** is rotated in the forward direction relative to the outer race **115**, for example. When the inner race **110** is rotated in the reverse direction relative to the outer race **115**, the inner race **110** and outer race **115** engage via the first rollers **130a** that are held between the roller support parts **121** and the first pocket parts **125a** in the circumferential direction. The operating mode of the positive clutch **100** is therefore the two-way lock mode that prohibits relative rotation of the inner race **110** and outer race **115** in both forward and reverse directions.

(69) A preset torque is determined by the friction between the sliding surfaces of the piston members **155** and the piston member accommodating holes **126**, and the resilience of the torque limiting member **150**. When torque exceeding this preset level is applied during forward rotation of the inner race **110** relative to the outer race **115**, the second rollers **130b** roll and move toward the bottom of the second pocket parts **125b** as shown in FIG. **16A**, pressing the piston members **155** and causing the torque limiting member **150** to radially expand through resilient deformation. As the second rollers **130b** roll out of the roller support parts **121** onto the outer circumferential surface of the inner race **110** as shown in FIG. **16B**, the inner race **110** and outer race **115** disengage and are freed from the torque transmission. This limits the magnitude of torque transmitted between the inner race **110** and the outer race **115**.

(70) As with the first embodiment, the angle of the load acting surface of the roller support part **121** and the inclination angle of the flat portion of the piston member **155** may be the same, or different. An angle difference between the load acting surface of the roller support part **121** and the inclination of the flat portion of the piston member **155** allows for change of the proportion of the torque limit load carried by the friction between the sliding surfaces of the piston members **155** and the piston member accommodating holes **126**, and the torque limit load carried by the resilience of the torque limiting member **150**. This helps improve performance stability.

(71) In this positive clutch **100**, when the switching member **170** is moved to the front side in the axial direction as shown in FIG. **17A**, and fixed in the second fixed position, only the second roller **130b** of each roller pair **131** is moved into the second pocket part **125b** by the action of the second pressing part **175** of the roller position changing part **172** as shown in FIG. **17B**. The second roller **130b** is thereby kept separated from the inner race **110**, and thus the operating mode of the positive clutch **100** is switched to the one-way lock mode that prohibits reverse rotation of the inner race **110** relative to the outer race **115**.

(72) Although not shown, the first pressing part **174** of the switching member **170** may be formed at a position on the front side in the axial direction relative to the second pressing part **175**, for example. It is then possible to hold only the first roller **130a** of each roller pair **131** separated from the inner race **110** by fixing the switching member **170** in the second fixed position. The positive clutch **100** can thus be switched to the one-way lock mode that prohibits forward rotation of the inner race **110** relative to the outer race **115**.

(73) When the switching member **170** is moved to the front side in the axial direction as shown in FIG. **18A**, and fixed in the third fixed position where the base part **171** of the switching member **170** makes contact with the rear face of the side plate **140**, the first roller **130a** of each roller pair **131** is moved into the first pocket part **125a** by the action of the first pressing part **174** of the roller position changing part **172** as shown in FIG. **18B**. The first rollers **130a** and second rollers **130b** are thus both held in the position separated from the inner race **110**. The operating mode of the positive

clutch **100** is thus switched to the two-way free mode that allows relative rotation of the inner race **110** and outer race **115** in both the forward and reverse directions.

(74) The positive clutch **100** according to the third embodiment provides the same effects as those of the positive clutch **100** of the first embodiment. Moreover, with the piston mechanism that uses the torque limiting member **150** employed as the torque limiter **151**, the clutch can achieve both compactness and performance stability.

(75) In the third embodiment described above, the torque limiter **151** has piston members each provided on the side in the forward direction of each second roller **130b** of each roller pair **131**. Alternatively, the torque limiter **151** may have piston members each provided on the side in the forward direction of each first roller **130a**, or the torque limiter **151** may have piston members each provided for each of the first rollers **130a** and second rollers **130b**. With piston members provided in the torque limiter **151** for both of the first rollers **130a** and second rollers **130b**, the torque limit load can be changed in each of the forward and reverse directions. This facilitates an improvement in performance stability, as well as provides a high degree of design freedom.

(76) In the third embodiment described above, the roller support parts **121** are formed on the outer circumferential surface of the inner race **110** while the first pocket parts **125a** and second pocket parts **125b** are formed on the inner circumferential surface of the outer race **115**. The torque limiting member **150** is fitted onto the outer race **115** so that its inner circumferential surface makes contact with the piston members **155** disposed in the outer race **115**. Instead, the roller support parts **121** may be formed on the inner circumferential surface of the outer race while the pocket parts may be formed on the outer circumferential surface of the inner race. The torque limiting member may be fitted in the inner race so that its outer circumferential surface makes contact with the piston members disposed in the inner race.

(77) In the third embodiment described above, the positive clutch **100** is configured to be switchable between a two-way lock mode, a one-way lock mode, and a two-way free mode. The positive clutch may be configured to be switchable between four operating modes including a one-way lock mode that prohibits forward rotation of the inner race relative to the outer race, and a one-way lock mode that prohibits reverse rotation of the inner race relative to the outer race.

(78) In the configuration described above, when switching the positive clutch between operating modes, the rollers **130** are separated from the roller support parts **121** by moving the switching member **170** in the axial direction independently of the rotation of the inner race **110** and outer race **115**. Alternatively, the switching member may be designed to separate the rollers from the roller support parts by rotating independently of the rotation of the inner race and outer race.

(79) While embodiments of the present invention have been described in detail, the present invention is not limited to the above-described embodiments and may be carried out with various design changes without departing from the scope of the present invention set forth in the claims.

(80) For example, while the inner race is fixed and the outer race is rotated in the configurations of the embodiments described above, other configurations are also possible, such as an inner race being rotated relative to a fixed outer race, or the inner race and outer race being both rotatable.

Claims

1. A positive clutch comprising: an inner race and an outer race that are coaxial and rotatable relative to each other; and a plurality of rollers disposed between the inner race and the outer race and transmitting or interrupting torque, the inner race and the outer race being configured to engage with each other via part or all of the plurality of rollers held in a circumferential direction between roller support parts and pocket parts, the roller support parts being formed on one of an outer circumferential surface of the inner race and an inner circumferential surface of the outer race, and the pocket parts being formed on the other of the outer circumferential surface of the inner race and the inner circumferential surface of the outer race and being configured to accommodate the rollers,

the positive clutch further comprising a torque limiting member that applies a radial load on the rollers to press the rollers against the roller support parts, the torque limiting member being coaxial with the inner race and the outer race, and being flat and annular in an axial direction, and moreover being resiliently deformable in a radial direction, wherein the torque limiting member is a spiral retaining ring.

2. The positive clutch according to claim 1, wherein the rollers each have an outer circumferential surface serving as a retaining ring receiving surface, and an outer circumferential surface or an inner circumferential surface of the torque limiting member is in contact with the retainer ring receiving surfaces of the plurality of rollers so as to apply a load to one or both axial ends of each of the plurality of rollers.

3. The positive clutch according to claim 1, wherein there is an angle difference between a wall surface of the roller support part and a wall surface of the pocket part that hold each of the rollers.

4. The positive clutch according to claim 1, wherein the plurality of rollers include first rollers held between the roller support parts and the pocket parts in the circumferential direction when one of the inner race and the outer race is rotated relative to the other in a forward direction, and second rollers held between the roller support parts and the pocket parts in the circumferential direction when one of the inner race and the outer race is rotated relative to the other in a reverse direction, the positive clutch further comprising: a biasing means that biases both the first rollers and the second rollers in the radial direction; a switching member that switches operating modes of the positive clutch by causing one or both of the first rollers and the second rollers to move into corresponding pocket parts against a biasing force of the biasing means; and piston members each having a flat portion for holding the first rollers between the first rollers and the roller support parts or the pocket parts, and/or piston members each having a flat portion for holding the second rollers between the second rollers and the roller support parts or the pocket parts, the torque limiting member having an outer circumferential surface or an inner circumferential surface in contact with the piston members.

5. A positive clutch comprising: an inner race and an outer race that are coaxial and rotatable relative to each other; and a plurality of rollers disposed between the inner race and the outer race and transmitting or interrupting torque, the inner race and the outer race being configured to engage with each other via part or all of the plurality of rollers held in a circumferential direction between roller support parts and pocket parts, the roller support parts being formed on one of an outer circumferential surface of the inner race and an inner circumferential surface of the outer race, and the pocket parts being formed on the other of the outer circumferential surface of the inner race and the inner circumferential surface of the outer race and being configured to accommodate the rollers, the positive clutch further comprising a torque limiting member that applies a radial load on the rollers to press the rollers against the roller support parts, the torque limiting member being coaxial with the inner race and the outer race, and being flat and annular in an axial direction, and moreover being resiliently deformable in a radial direction, wherein the rollers each have an outer circumferential surface serving as a retaining ring receiving surface, and an outer circumferential surface or an inner circumferential surface of the torque limiting member is in contact with the retainer ring receiving surfaces of the plurality of rollers so as to apply a load to one or both axial ends of each of the plurality of rollers.

6. A positive clutch comprising: an inner race and an outer race that are coaxial and rotatable relative to each other; and a plurality of rollers disposed between the inner race and the outer race and transmitting or interrupting torque, the inner race and the outer race being configured to engage with each other via part or all of the plurality of rollers held in a circumferential direction between roller support parts and pocket parts, the roller support parts being formed on one of an outer circumferential surface of the inner race and an inner circumferential surface of the outer race, and the pocket parts being formed on the other of the outer circumferential surface of the inner race and the inner circumferential surface of the outer race and being configured to accommodate the rollers,

the positive clutch further comprising a torque limiting member that applies a radial load on the rollers to press the rollers against the roller support parts, the torque limiting member being coaxial with the inner race and the outer race, and being flat and annular in an axial direction, and moreover being resiliently deformable in a radial direction, wherein there is an angle difference between a wall surface of the roller support part and a wall surface of the pocket part that hold each of the rollers.
